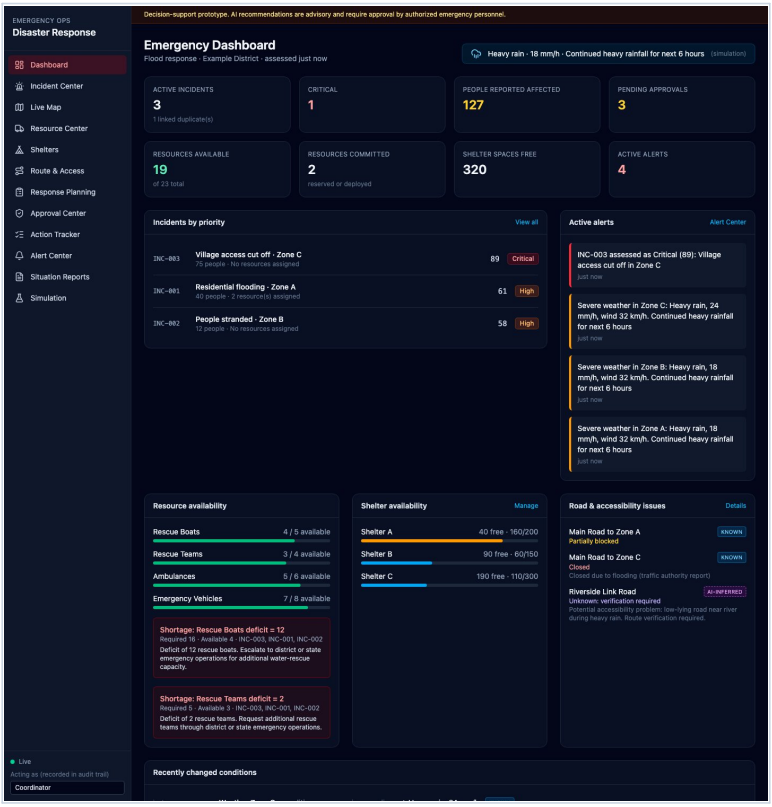
USER MANUAL

Disaster Response Coordination Console

Agentic AI for emergency incident assessment, resource planning and human-approved response



The Emergency Dashboard: the first screen you see.

For	Emergency coordinators, duty officers and response staff who use the console
Covers	Raising incidents · what the AI agents do automatically · every tab and button · human approvals · reports
Version	1.0 · 09 October 2026 · application POC v0.1

Safety: This system **recommends**; it never dispatches anything. Every recommendation waits for an authorized coordinator. Resources are reserved only when a person clicks **Approve**.

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1. About this application

The Disaster Response Coordination Console helps an emergency coordination centre handle many incidents at once during a disaster such as a flood, landslide or urban fire. Reports arrive from phone calls, field teams, sensors, government alerts and citizens. The console turns them into a single, prioritised operational picture and proposes how to use the available boats, rescue teams, ambulances and vehicles.

Behind the screens, **eight AI agents** work together every time something new happens. They read reports, check the weather and the roads, look at which resources are free, score each incident, build alternative plans and double-check them. The result is a set of **recommendations** that a human coordinator approves, changes or rejects.

1.1 The four golden rules

Rule	What it means for you
AI recommends, you decide	Nothing is dispatched or reserved automatically. Every recommendation appears as a *pending* action in the Approval Center.
Every fact shows where it came from	Badges tell you whether a value is **Known** (official registry or API data), **Reported** (from a report, not yet verified) or **AI-Inferred** (derived by the AI or a rule).
Nothing is invented	If something is unknown (people count, location, road status, weather), the system shows **Verification Required** instead of guessing. It never invents resources, roads or numbers.
Scores come from fixed rules	Priority scores, shortages and allocations are calculated by transparent, tested rules, not by the AI model. The AI only reads text and writes explanations.

1.2 Key terms

Term	Meaning
Incident	One emergency situation, e.g. *INC-003 Village access cut off*. Has a type, zone, location, people affected, needs and a priority.
Raw report	A message exactly as received (call transcript, citizen text, sensor event). The Intake agent turns it into an incident.
Priority	A score from 0 to 100 with a level: Critical, High, Moderate, Low or Verification Required.
Resource	A response unit: rescue boat (BOAT-), rescue team (TEAM-), ambulance (AMB-), emergency vehicle (VEH-), medical team (MED-).
Plan	One complete proposal for using the resources. Three are built each time: A Priority-first, B Coverage, C Verified access only.
Action	One recommended task (e.g. *deploy BOAT-002 and TEAM-002 to INC-003*). Moves through Pending → Approved → In progress → Completed.
Shortage	Resources needed minus resources available, with the exact deficit and an escalation suggestion.
Run	One automatic pass of all eight agents (RUN-001, RUN-002 …). Each run is recorded with a full trace.
Alert	A notification about something important: critical incident, road closed, shelter nearly full, weather worsening, overdue action.

1.3 Colours and badges

Badge / colour	Meaning
Critical (red)	Score 75 or more: act first
High (orange)	Score 55–74
Moderate (yellow)	Score 35–54
Low (green)	Score below 35
Verification Required (violet)	Location unknown, or an unconfirmed low-confidence report: confirm before acting
KNOWN (blue badge)	Official or registry data, e.g. a road closure from the traffic authority
REPORTED (grey badge)	Stated in a report but not yet verified
AI-INFERRED (dashed violet badge)	Derived by the AI or a rule; treat as a lead, not a fact

2. Getting started

Open the console in your web browser (by default **http://localhost:5173**). The screen has three parts:

- **Sidebar (left):** the 13 tabs, the connection indicator and your name.
- **Safety banner (top, amber):** a permanent reminder that recommendations need approval.
- **Main area:** the selected tab.

2.1 Before you start working

1. Check the dot at the bottom of the sidebar. **Live** (green) means the screen updates by itself. **Offline** (red) means the server cannot be reached; the screen retries automatically.
2. Type your name in **Acting as** (bottom of the sidebar). Every approval, rejection and comment is recorded in the audit trail under this name.
3. Open the **Dashboard** to see the current situation.

> **Note:** You never need to refresh the page. When the agents finish a run, or anyone changes something, every open tab reloads its data within a second and new alerts pop up as notifications in the bottom-right corner.

2.2 The tabs at a glance

Tab	Use it to…
Dashboard	See the whole situation: key numbers, incidents by priority, alerts, resources, shelters, roads, recent changes.
Incident Center	Register a new incident; search and filter all incidents.
Incident Details	(open any incident) See the score breakdown, report, weather, routes, actions; review duplicates; verify.
Live Map	See incidents, resources, shelters and roads on a map; click anything for details.
Resource Center	See every resource and its status; mark units unavailable or back in service; view history.
Shelters	See capacity and free space; update occupancy and status.
Route & Access	See which zones are reachable by road; known closures vs AI-inferred problems; update road status.
Response Planning	Compare Plans A, B and C; see shortages, reviewer findings and the full agent trace; run an assessment.
Approval Center	Decide on recommendations: approve, reject, modify resources, request reassessment, comment.
Action Tracker	Follow actions from approval to completion; start and complete deployments; add remarks.
Alert Center	Review, filter and acknowledge alerts.
Situation Reports	Generate and download PDF situation reports.
Simulation	Training and demo controls: load test scenarios, inject reports, close roads, worsen weather.

3. The complete workflow at a glance

Everything in the console follows one cycle. You (blue) raise or receive an incident; the AI agents (purple) analyse it and propose plans automatically; a coordinator (orange) approves; the team executes and you track it (green). Whenever conditions change, the cycle repeats automatically for the affected incidents.

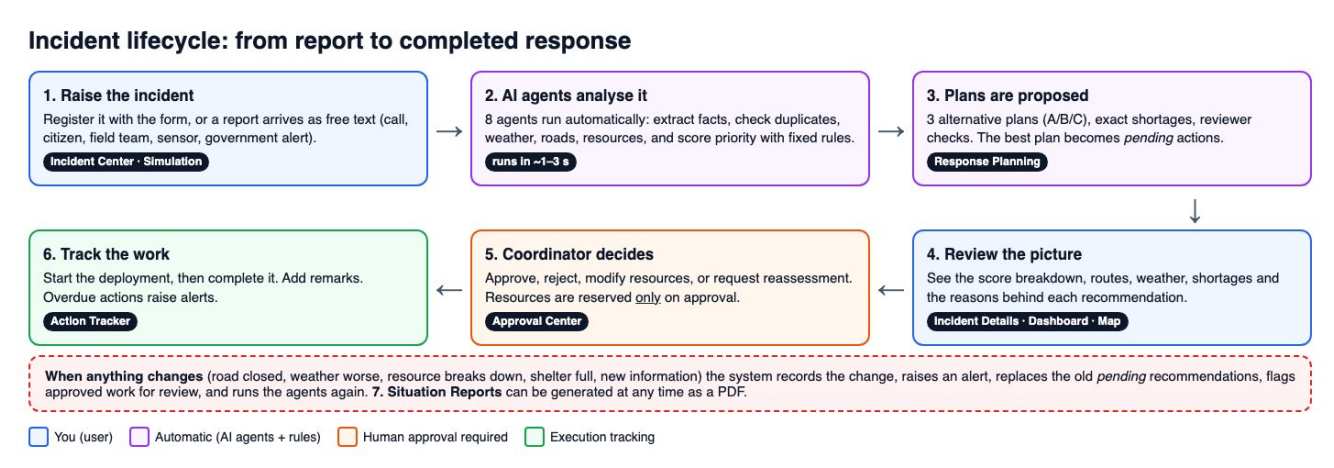


Figure 3.1: Incident lifecycle and where each step appears in the console.

Step	Who	Where you see it	Typical time
1. Raise the incident	You (or an incoming report)	Incident Center · Simulation	1 minute to fill in
2. Agents analyse	Automatic	Incident Details (score), Response Planning (trace)	1–3 seconds
3. Plans proposed	Automatic	Response Planning, Approval Center	same run
4. Review	You	Incident Details, Dashboard, Live Map, Route & Access	as needed
5. Decide	Coordinator	Approval Center	before the due time shown on the card
6. Execute and track	Coordinator / field teams	Action Tracker, Resource Center	until completed
7. Report	You	Situation Reports	seconds

4. Step 1: Raising an incident

There are two ways an incident enters the system. Both lead to exactly the same automatic analysis.

	Method A: Register it yourself	Method B: A report arrives as text or data
When	You already know the facts (e.g. a field team phoned you)	A message comes in from a call, citizen, field team, sensor or government feed
Where	Incident Center → **New incident**	Arrives automatically through the intake interface (in training: Simulation → **Inject a source report**)
Who fills the fields	You, in a form	The Intake agent extracts them with AI
Result	A new incident (INC-…) immediately; score within seconds	A new incident (INC-…) within seconds, linked to duplicates if any

4.1 Method A: Register a new incident (step by step)

1. Click **Incident Center** in the sidebar.
2. Click the red **New incident** button (top right). The *Register new incident* form opens.
3. Fill in the fields (see the table below). Only **Title** and **Description** are mandatory; choose a **Zone** to fill the coordinates automatically.
4. Leave **People affected** empty if you do not know it. The system will mark it *Verification required* instead of guessing.
5. Click **Register incident**. You are taken to the new incident's page. It first shows *Not yet assessed*; a few seconds later the score appears by itself.

Field	What to enter	Example (used throughout this manual)
Incident type	flood, cyclone, earthquake, wildfire, landslide, industrial accident, urban emergency	flood
Source	emergency call, field team, IoT sensor, government alert, citizen report	field team
Title *	A short name	Families trapped in flooded school
Description *	What is happening, in the reporter's words	Water is rising around the Government School on River Street. About 30 people are trapped on the first floor, two children are injured. The approach road is under water.
Zone	Pick from the list (fills latitude/longitude)	Zone D
Latitude / Longitude	Adjust if you know the exact spot	13.0660, 80.2360
Reported at	When the report was made (defaults to now)	now
People affected	A whole number, or empty if unknown	30
Requested assistance	Click the chips that apply	water rescue, medical check
Supporting information	Caller details, landmarks, photos received	(optional)

> **Tip:** Write the description as it was reported. The AI reads it to recognise threats such as *trapped*, *injured* or *stranded*, which raise the priority. You do not need to choose these yourself.

4.2 Method B: A report arrives as free text

Reports from outside systems (call-centre transcripts, citizen apps, sensor feeds, government alerts) are sent to the console's intake interface and processed by the **Incident Intake agent**. For training, use **Simulation → Inject a source report**: choose the source, edit the text, and click **Send to Intake agent**. Example citizen message:

"water rising fast near the school on river street zone D, people stuck upstairs, kids hurt, please send boat"

In our example the AI extracted: type **flood**, zone **Zone D**, needs **water rescue**, threats **stranded**; the people count stayed **unknown** because the message gives no number. It also recognised the message as a **possible duplicate** of INC-005 (score 0.79), so it was created as INC-006 and linked for your review rather than counted twice.

4.3 What you will see in the first seconds

Time	What happens	Where it shows
0 s	Incident saved with a new ID (INC-005)	You land on Incident Details
~0.5 s	AI classifies threats from the description; duplicate check runs	Incident Details → Coordinator controls
1–3 s	The eight agents run (next chapter)	Response Planning → agent trace
~3 s	Score and level appear; new pending action(s) created; alerts raised if critical	Incident Details, Approval Center, Dashboard, notifications

5. What happens automatically: the eight AI agents

Each new incident, new report, changed condition or click on **Run assessment** starts one **run**. In a run, eight specialised agents each do one job and pass their results on through a shared workspace. Three of them work at the same time to save time. Only two of the eight use the AI language model; the rest use fixed, tested rules.

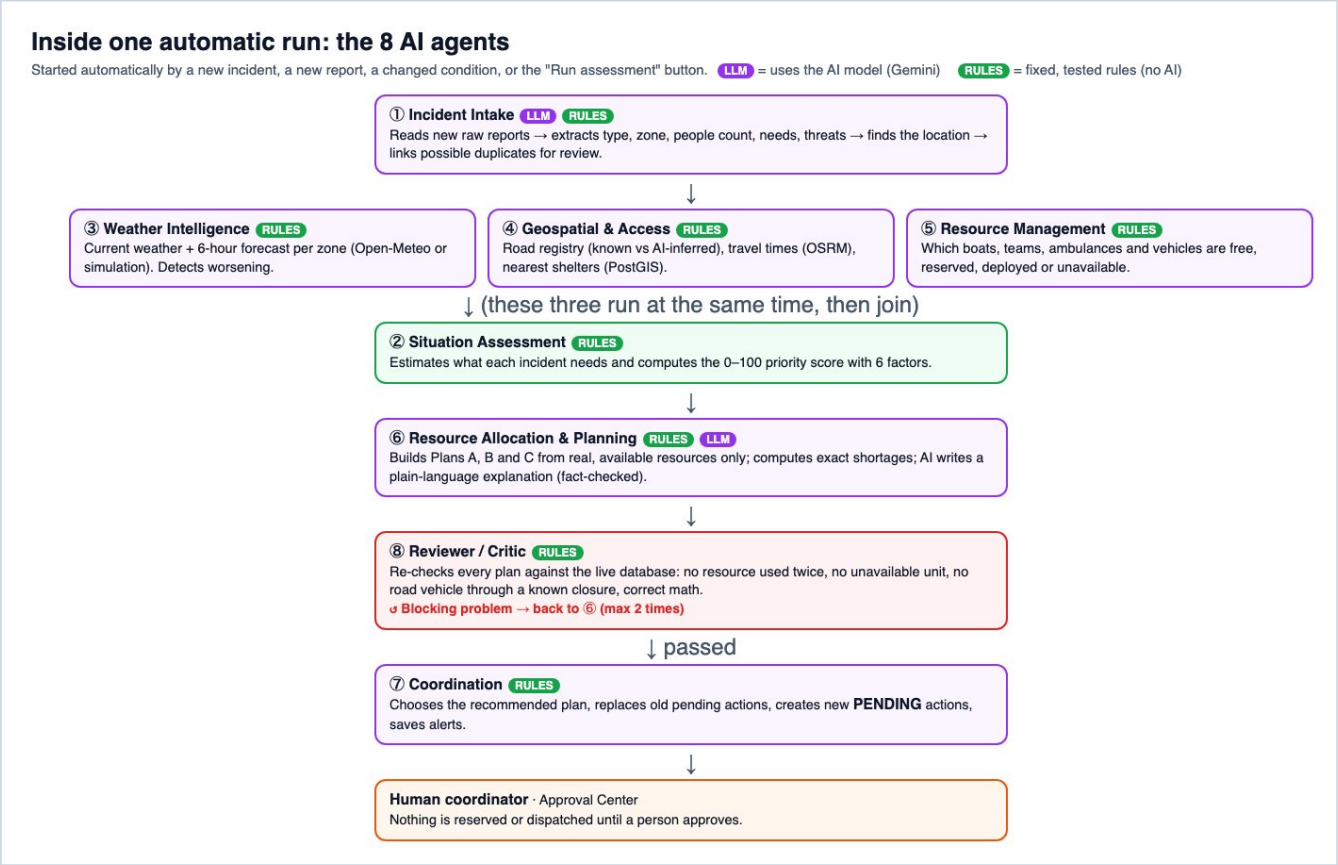


Figure 5.1: Inside one automatic run.

5.1 What each agent does, in plain language

Agent	What it does	Uses AI?	Where you see the result
1 Incident Intake	Reads new reports; extracts type, zone, people count, needs, threats and infrastructure problems; finds the location; links possible duplicates.	Yes (reading text)	Incident Center, Incident Details
3 Weather Intelligence	Gets current weather and the 6-hour forecast for every affected zone; notices when it gets worse.	No	Dashboard banner, Incident Details → Zone weather, alerts
4 Geospatial and Access	Checks the road registry for each zone (known closures vs AI-inferred problems), travel times from each base, nearest shelters.	No	Route & Access, Live Map
5 Resource Management	Lists every resource and whether it is free, reserved, deployed, returning or unavailable.	No	Resource Center, Dashboard
2 Situation Assessment	Estimates what each incident needs and calculates its 0–100 priority score.	No	Incident Details → Priority assessment
6 Allocation and Planning	Builds Plans A, B and C from real, free resources only; calculates exact shortages; the AI writes a plain-language explanation that is fact-checked.	Yes (explanation only)	Response Planning
8 Reviewer / Critic	Re-checks every plan against the live data: no resource twice, no unavailable unit, no road vehicle through a known closure, correct maths. Sends the plan back if needed.	No	Response Planning → Reviewer findings
7 Coordination	Picks the recommended plan, replaces older pending recommendations, creates new pending actions, saves alerts.	No	Approval Center, Action Tracker, Alert Center

5.2 A real run, step by step

This is the actual trace of **RUN-003**, started when the citizen message from section 4.2 arrived (Response Planning → *Agent trace* shows exactly this):

Agent	Time	AI used	Result
Incident Intake	1638 ms	Yes	Created 1 incident(s) ['INC-006']; 1 duplicate link(s); 0 report(s) need manual review; 6 active incident(s)
Resource Management	4 ms	No	Available: ambulance 5, emergency_vehicle 7, rescue_boat 4, rescue_team 3; shelters at/above threshold: none
Weather Intelligence	19 ms	No	Weather for 3 zone(s); worsening in 0; no data for ['Zone D']
Geospatial Accessibility	196 ms	No	Access: Zone A partially_blocked (known), Zone B open (known), Zone C closed (known), Zone D unknown (reported); isolated by road: ['Zone C']
Situation Assessment	10 ms	No	INC-003 89 critical; INC-001 61 high; INC-005 61 high; INC-002 58 high; INC-004 50 moderate; INC-006 47 verification_required
Resource Allocation	1375 ms	Yes	Plans A/B/C generated; recommended B (score 31.5); shortages: rescue_boat -16, rescue_team -4, ambulance -1
Reviewer Critic	4 ms	No	Iteration 1: 0 blocking, 6 warning, 3 info finding(s) across plans
Emergency Coordination	16 ms	No	6 active incident(s); critical: INC-003. Recommended PLAN-008 (Plan B: Coverage) with 4 action(s) pending coordinator approval. Shortages: rescue_boat

Note: Each row can be expanded in the console to see every tool the agent used, for example *llm.extract_incident → type=flood zone=Zone D people=None*, *duplicates.find_candidates → INC-005 potential_duplicate 0.79*, *osrm.table → 114 routed*, *postgis.nearest_shelters → SHL-A 1341 m*.

6. How information is extracted from reports

When a report arrives as text, the Intake agent asks the AI model (Gemini) to fill a fixed form. The AI may only choose from approved labels and must leave a field empty when the report does not say it. The system then checks the AI's answer with its own rules.

Extracted field	How it is decided	Shown as
Incident type	From words such as flood, landslide, fire, gas leak	Type on every incident
Zone and location	Coordinates in the report (Reported) → a named zone or known place name such as "river street" (AI-Inferred, approximate) → otherwise **Location unknown**	Coordinates with a badge; unknown locations are never drawn on the map
People affected	Only a stated number ("about 30 people" → 30). Ranges (10–15), "lots of people", families or houses are **not** turned into a number.	Number, or *Unknown: verification required*
Threats to life	trapped, stranded, injured, water rescue, isolated / cut off, evacuation needed, property damage only, fire	Priority factor *Threat to life* (AI-Inferred)
Assistance needed	water rescue, evacuation, medical check, first aid, fire suppression, search and rescue, shelter …	Chips on the incident
Infrastructure problems	main road closed, road partially flooded, road blocked, power outage, bridge damaged …	Chips and the *Infrastructure impact* factor
Confidence	Field team or government alert with type and zone → high; emergency call → medium; citizen report or sensor → low	Status panel on Incident Details

Important: If the AI service is unavailable, the system automatically switches to a built-in rule-based reader. Work continues, but the results are marked confidence **low** and the agent trace shows *fallback*.

6.1 Duplicate detection

Every new incident is compared with existing ones using four measures: distance between locations (within 500 m), time between reports (within 60 minutes), same type, and how similar the descriptions are. The weighted result is a score from 0 to 1:

- **0.70 or more:** *possible duplicate*. Linked for your review. The duplicate is not given its own resources, so the same people are not counted twice.
- **0.50–0.69:** *related*. Shown for information only.
- **Nothing is ever merged or deleted automatically.** You confirm or reject the link on the Incident Details page.

Example: INC-006 vs INC-005 scored **0.79**: distance 155 m (0.69), time 0 min (1.00), same type, text similarity 0.61.

7. How priority is decided

Every incident gets a score from 0 to 100 made of six factors. The rules are fixed and the same for everyone; the AI never sets or changes a score. Incident Details shows each factor, the points it earned and where its input came from.

Factor	Max	How points are given
Threat to life	25	trapped / stranded / injured / water rescue 22 · isolated / cut off 18 · evacuation needed 12 · property damage only 5 · no information 0 (and *Confirm threat to life*)
People affected	20	unknown 0 (and *Confirm number of people affected*) · 1–9: 5 · 10–24: 8 · 25–49: 14 · 50 or more: 20
Access route	15	best **known** road to the zone: open 0 · partially blocked 8 · closed 15 · unknown or only AI-inferred 10 (and *Route verification required*)
Weather / escalation	15	severe warning and heavy rain forecast 13 · heavy rain 10 · moderate rain 5 · none 0 · +2 if wind is 50 km/h or more · no data 0 (and *Weather data unavailable*)
Infrastructure impact	10	main road closed 8 · road partially flooded or power/utility loss 6 · other reported damage 4 · none 0
Response gap	15	nothing assigned 15 · assigned but less than needed 8 · fully covered 0

Levels: Critical 75 or more · High 55–74 · Moderate 35–54 · Low below 35. The level becomes **Verification Required** (the score is still shown) when the location is unknown, or when a low-confidence, unverified report is not confirmed by a verified duplicate.

Example (INC-005, the flooded school): 61 / 100, High. Threat to life 22 (trapped, injured), people 14 (30 reported), access 10 (no confirmed road to Zone D), weather 0 (no data for Zone D, so nothing is assumed), infrastructure 0, response gap 15 (nothing assigned).

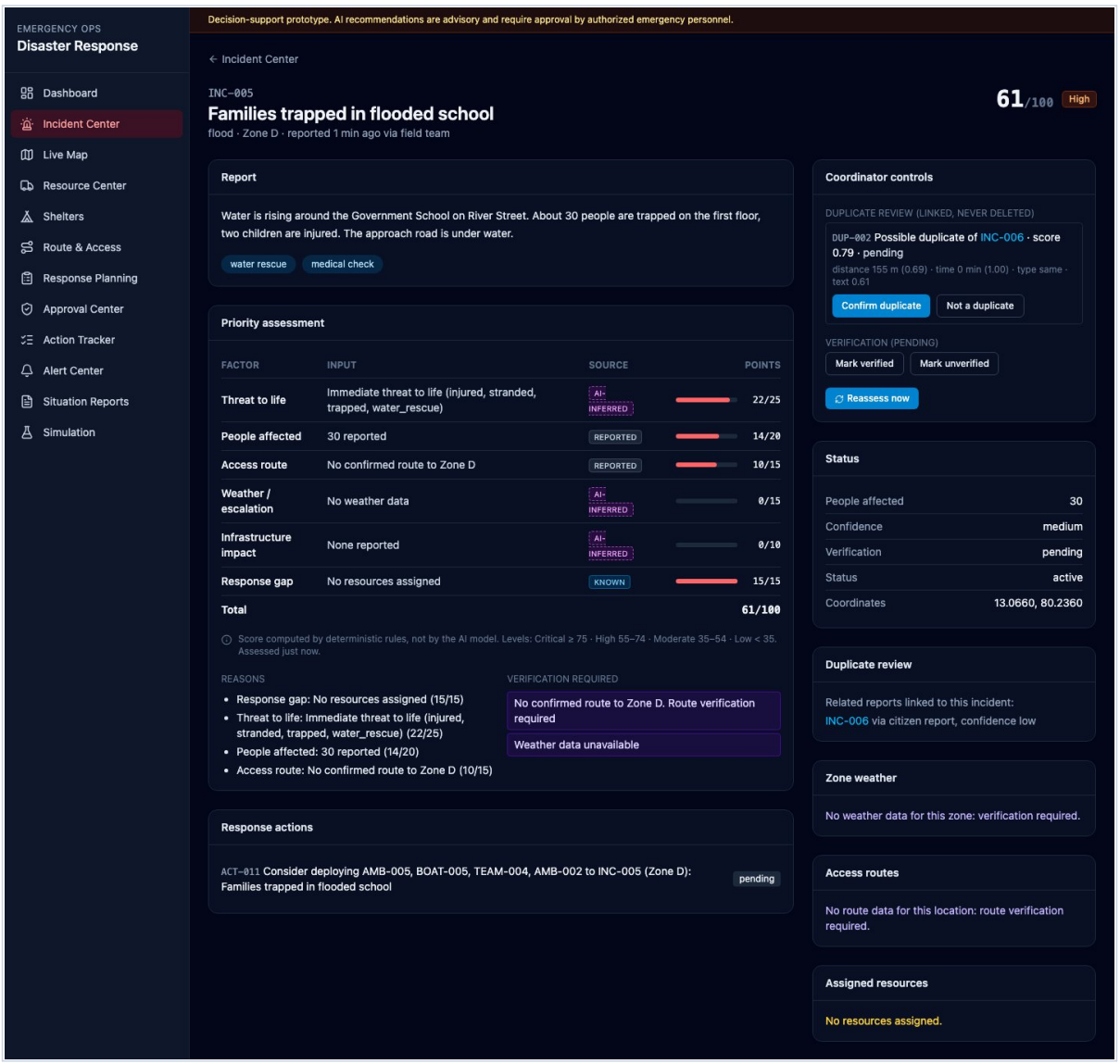


Figure 7.1: Incident Details for INC-005 with the factor-by-factor score, sources and verification items.

8. How resources are matched and plans are built

After scoring, the Allocation agent decides **what each incident needs** and **which free resources could meet it**. Only resources with status *available* are used, and no resource can appear twice in a plan.

8.1 What an incident needs

Situation	Need (n = people affected)
Water rescue (flood with closed or partly blocked roads, or people stranded)	Rescue boats = n ÷ 8 rounded up · Rescue teams = 1 + one more per 30 people
Evacuation by road	Emergency vehicles = n ÷ 10 rounded up · Rescue teams as above
People trapped (landslide, collapse, fire)	Rescue teams for search and rescue
Medical help requested or people injured	Ambulances = 5% of n rounded up, at least 1
Fire	Reported as an **external need** ("request fire services"); the registry has no fire units and none are invented
People count unknown	Minimum of 1 unit of each needed type, marked *verification required*

8.2 Which resource is chosen

- **Can it get there?** Road-only units (ambulances, vehicles) are never sent to a zone whose known roads are all closed. Boats may go, but always with the note *"Water and route conditions permit boat deployment: verification required"*.
- **Is it the right kind?** Water-rescue capable units first when boats are needed.
- **How fast?** Shortest travel time (live road routing, or a clearly labelled estimate).
- **How big?** Larger capacity first when times are equal.

8.3 Three alternative plans

Plan	Strategy	Good when
A Priority-first	Give the highest-priority incident everything it needs before moving to the next	One incident is far more serious than all others
B Coverage	First give every Critical and High incident a minimum response, then top up in priority order	Resources are scarce and many incidents are serious
C Verified access only	Allocate only where road access is confirmed and the report needs no verification	Information is uncertain and you want to act only on confirmed facts

The plan that covers the most priority-weighted need is marked **Recommended**, but only if the Reviewer found no blocking problem. Each plan also lists **shortages** with the exact deficit, for example *Rescue boats: 20 required, 4 available, deficit 16. Escalate to district or state emergency operations*. The system never pretends to have resources it does not have.

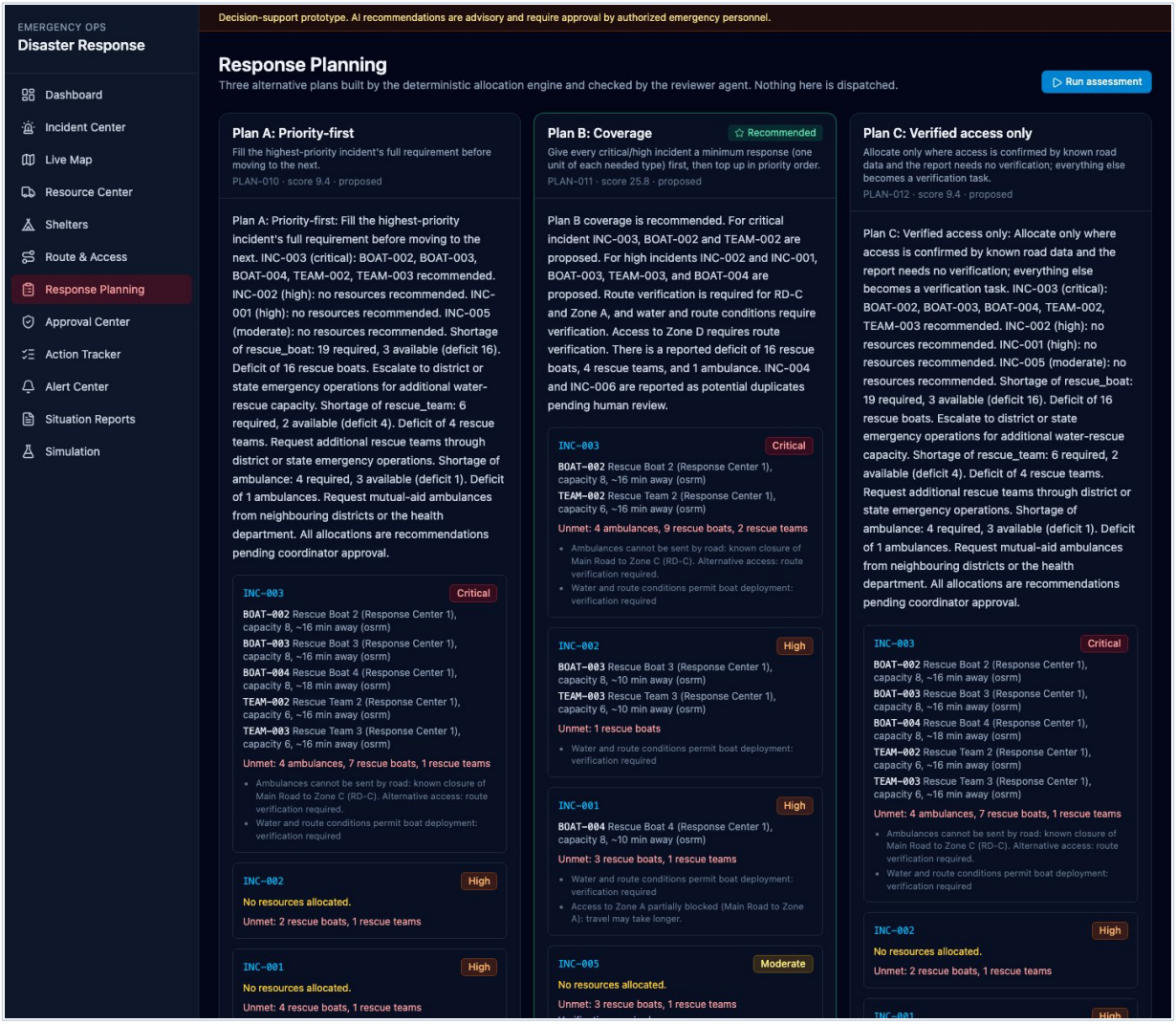


Figure 8.1: Response Planning with Plans A, B (Recommended) and C side by side.

9. Where human approval is needed

The AI agents recommend continuously, but these decisions belong to people. None of them happens automatically.

#	Decision	Where	What happens when you decide
1	**Approve** a recommended action	Approval Center	Resources become **Reserved**. This is the only way resources are committed. If a resource was taken meanwhile, you get a clear message and a fresh plan is prepared.
2	**Reject** an action	Approval Center	The action is closed with your comment; resources stay free.
3	**Modify resources**	Approval Center	Choose other free resources. Units that cannot reach the zone are refused with the reason.
4	**Request reassessment**	Approval Center	All agents run again with the latest data.
5	**Start** / **Complete** a deployment	Action Tracker	Resources move to Deployed, then Returning.
6	**Confirm or reject a duplicate**	Incident Details → Coordinator controls	Decides whether two reports describe the same event (nothing is deleted).
7	**Verify** an incident	Incident Details	Marks the report verified, which can lift a *Verification Required* level.
8	**Review approved work after a change**	Action Tracker (audit note)	Approved actions are never changed automatically; they receive a *coordinator review required* note.
9	**Return a resource to service**	Resource Center	Returning or unavailable → Available.

Safety: The AI never: dispatches a unit, reserves a resource, deletes or merges a report, changes an approved decision, or makes up a number, road or resource.

10. Tab-by-tab guide

Each section shows the tab as it looks during the worked example (chapter 11), what it displays and what you can do there.

10.1 Dashboard

Your starting point: the whole situation on one screen.

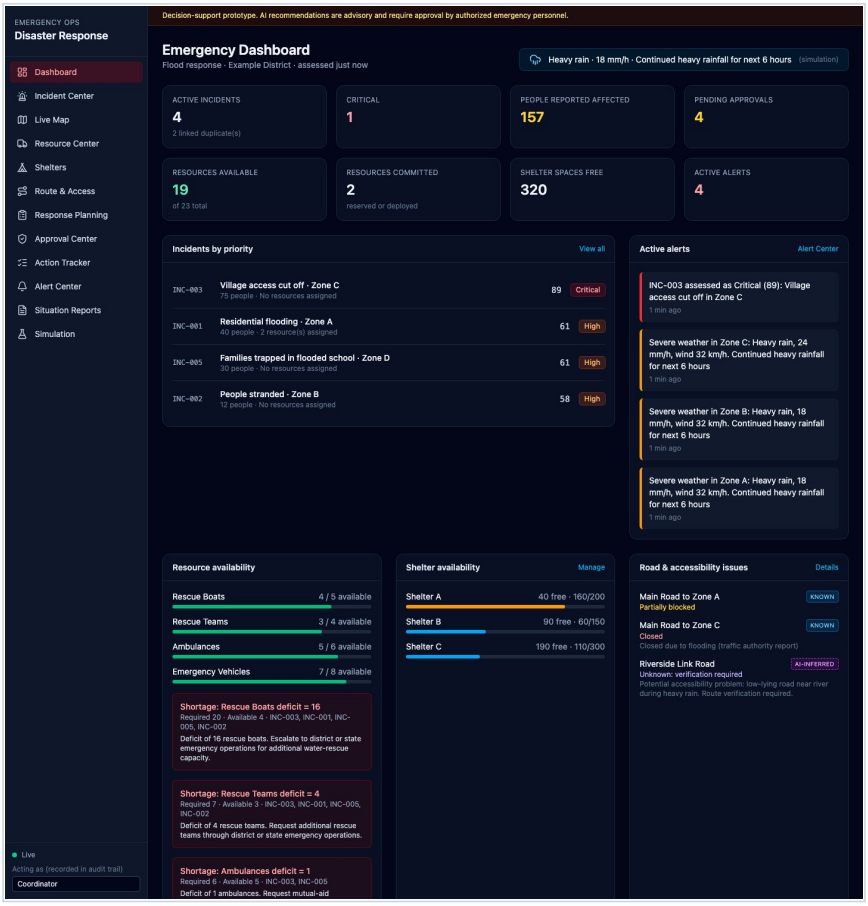


Figure 10.1: Emergency Dashboard.

What you see

- **Eight key figures:** active incidents (duplicates excluded), critical incidents, people reported affected (and how many incidents have an unknown count), pending approvals, resources available and committed, free shelter spaces, active alerts.
- **Incidents by priority:** highest score first, with people affected and whether resources are assigned.
- **Active alerts**, **Resource availability** per type with any **shortage**, **Shelter availability** bars, **Road and accessibility issues** with provenance badges.
- **Weather banner** (top right) and **Recently changed conditions** (old value crossed out → new value).

What you can do

- Click an incident to open its details; use *View all*, *Alert Center*, *Manage* and *Details* links to jump to other tabs.

10.2 Incident Center

Register incidents and find any incident quickly.

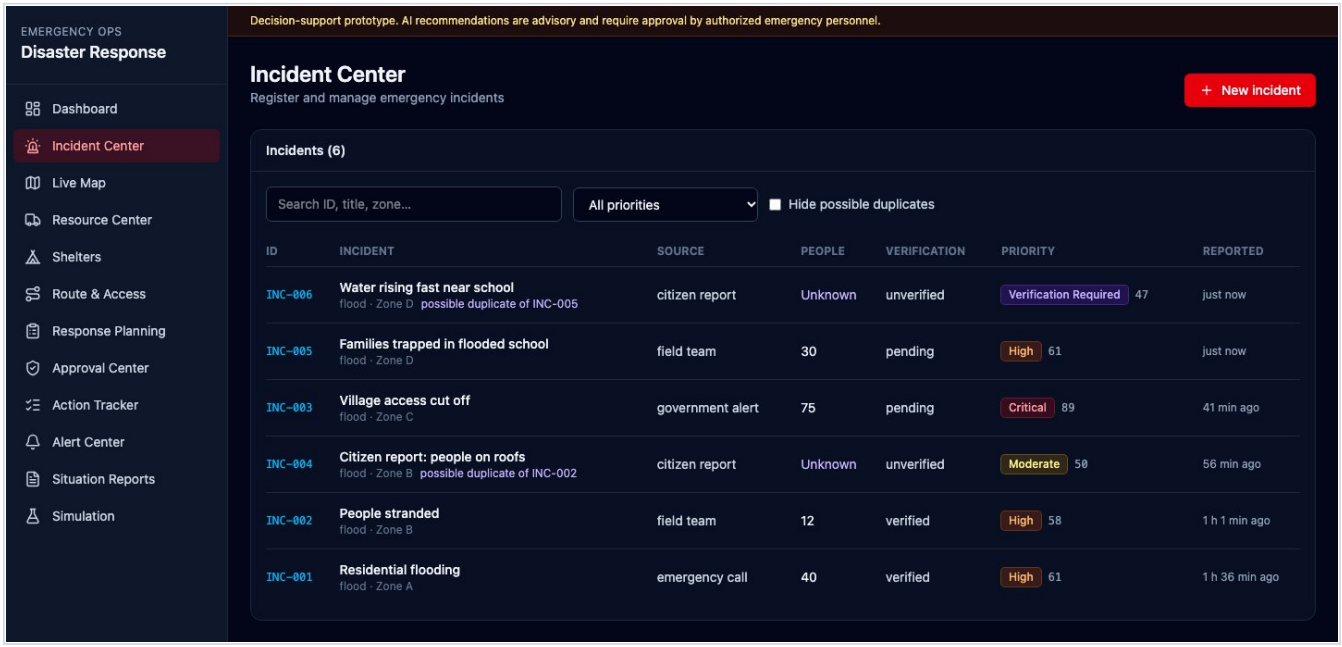


Figure 10.2: Incident Center list with priority, verification and duplicate notes.

What you see

- A table of all incidents with ID, title, type, zone, source, people affected (or *Unknown*), verification status, priority badge and score, and when it was reported.
- Possible duplicates are marked *possible duplicate of INC-…*.

What you can do

- **New incident**: register an incident (chapter 4).
- **Search** by ID, title, zone or description; **filter** by priority level or *Not assessed*; tick **Hide possible duplicates**.
- Click an ID to open Incident Details.

10.3 Incident Details

Everything about one incident, and the coordinator's controls for it.

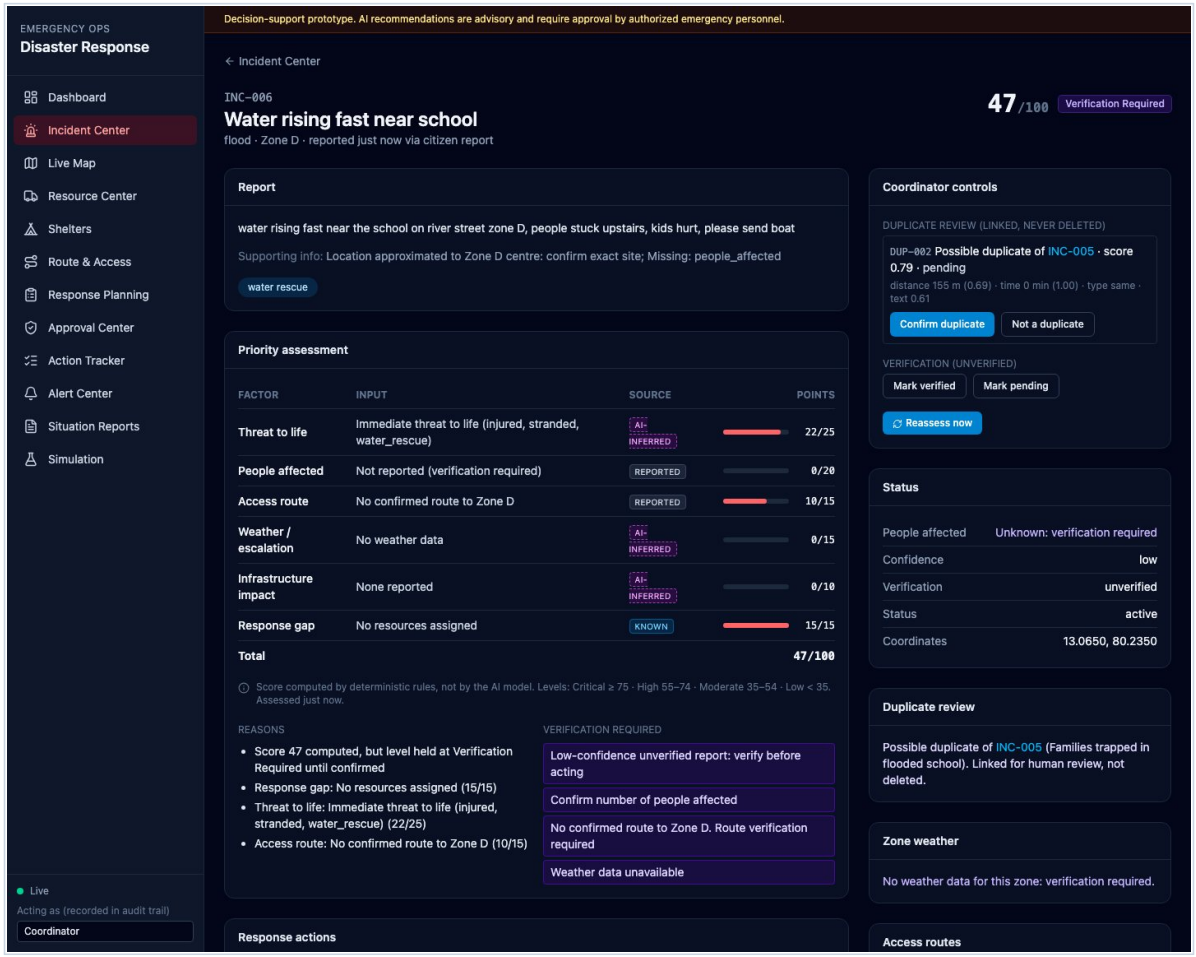


Figure 10.3: Incident Details for INC-006 (a citizen report): duplicate review, unknown people count, Verification Required.

What you see

- **Report** text with assistance and infrastructure chips.
- **Priority assessment**: each factor's input, source badge and points; reasons; **Verification required** items (see Figure 7.1).
- **Response actions** for this incident and their status.
- Right column: **Coordinator controls**, **Status** (people, confidence, verification, coordinates), **Duplicate review**, **Zone weather**, **Access routes**, **Assigned resources**.

What you can do

- **Confirm duplicate** / **Not a duplicate** for each link, with the full score breakdown (distance, time, type, text).
- **Mark verified / unverified / pending**.
- **Reassess now**: runs all agents again.

> **Tip:** A *Verification Required* level is not an error. It means the system will not act on unconfirmed or unknown facts until a person checks them.

10.4 Live Map

See where everything is.

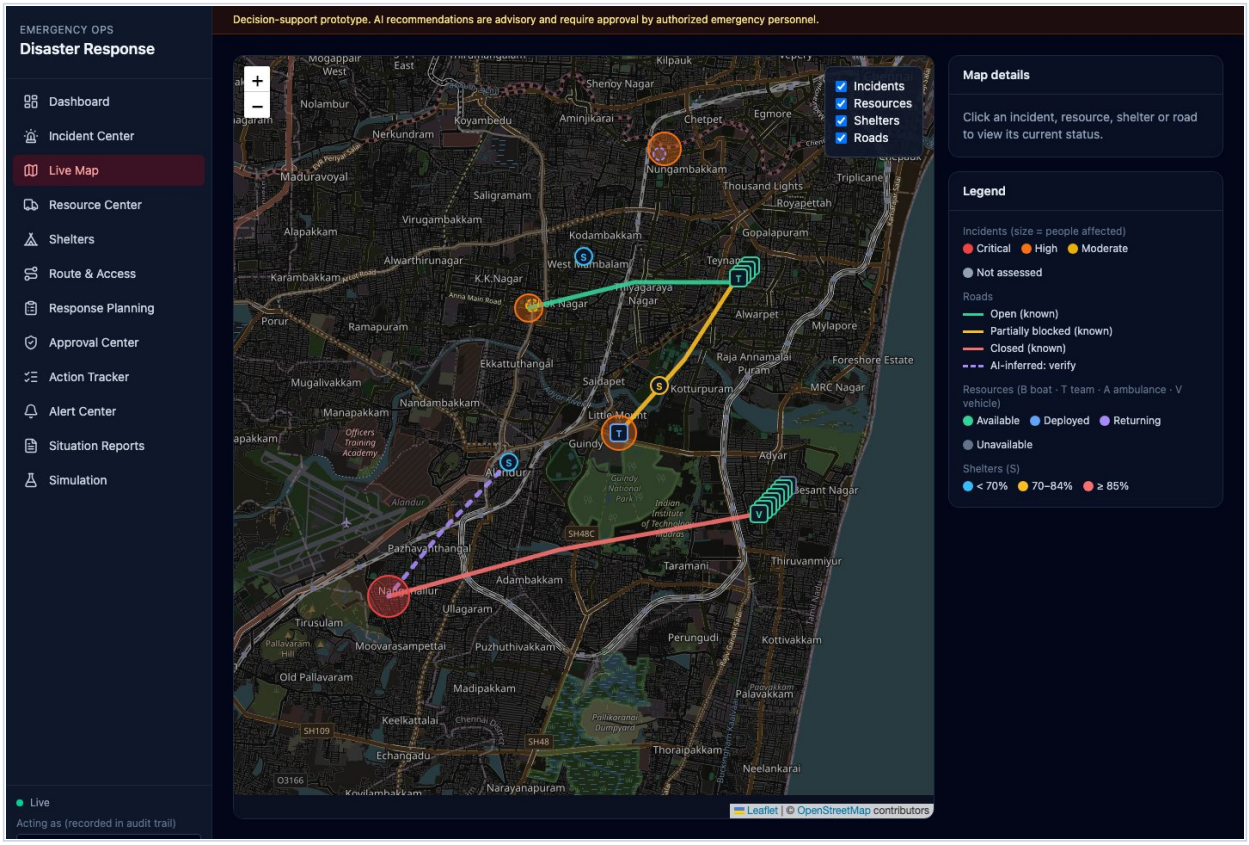


Figure 10.4: Live Map.

What you see

- Incidents as circles coloured by priority (size grows with people affected; duplicates are dashed).
- Resources as small lettered squares (B boat, T team, A ambulance, V vehicle, M medical) coloured by status.
- Shelters coloured by how full they are; roads coloured by status (open, partially blocked, closed, unknown).
- A legend and layer switches; incidents with an unknown location are listed but never drawn at a guessed point.

What you can do

- Click any marker or road to see its details in the side panel.
- Turn layers on or off.

10.5 Resource Center

The registry of every response unit.

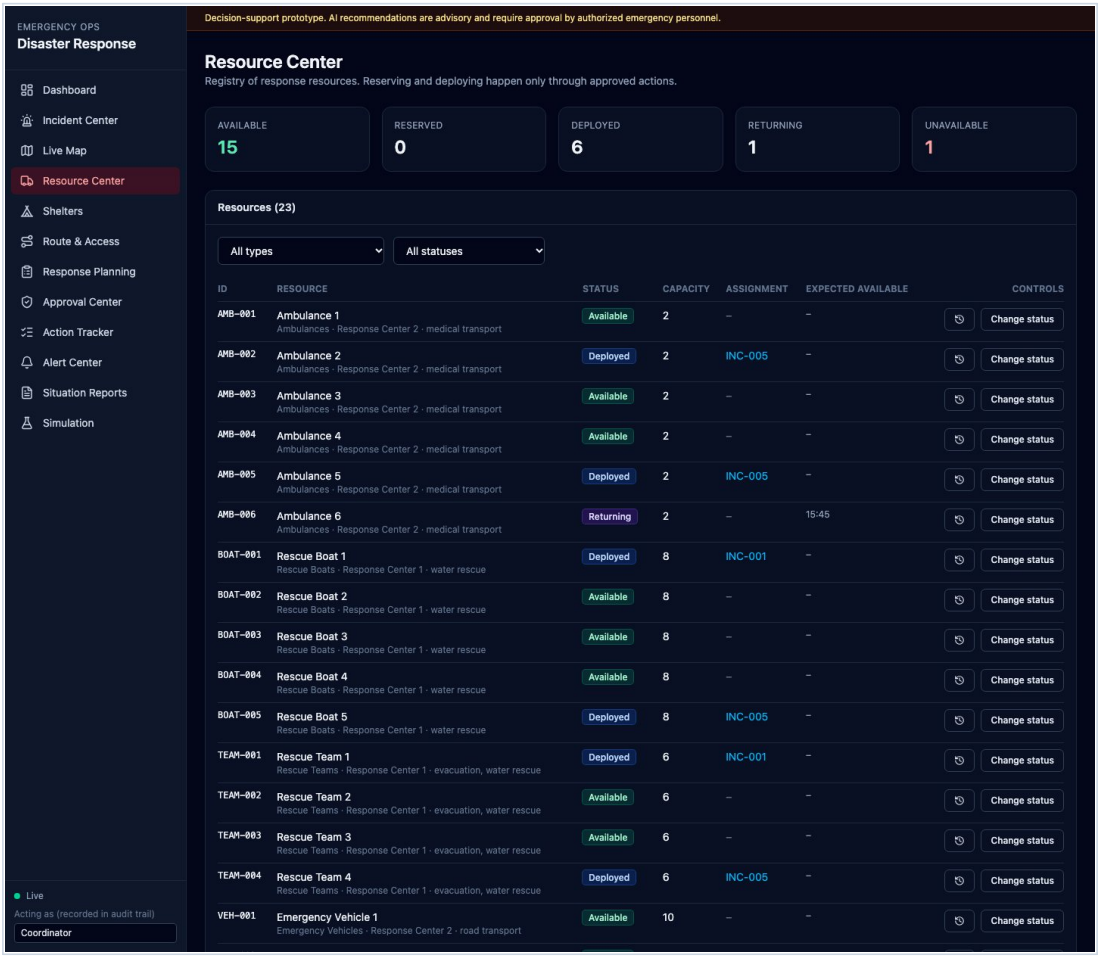


Figure 10.5: Resource Center after ACT-011 was approved and started.

What you see

- Counts by status: Available, Reserved, Deployed, Returning, Unavailable.
- Every resource with type, base, capabilities, status badge, capacity, the incident it is assigned to and its expected availability.

What you can do

- **Filter** by type and status.
- **Change status**: only valid changes are offered (e.g. available → unavailable with a required reason and optional "back in N hours"; returning → available). Reserving and deploying happen only through approved actions.
- **History** (clock icon): every status change with time, reason and who made it.

> **Tip:** Marking an assigned unit unavailable raises an alert, flags its action for review and recalculates the plan automatically.

10.6 Shelters

Capacity and occupancy of every shelter.

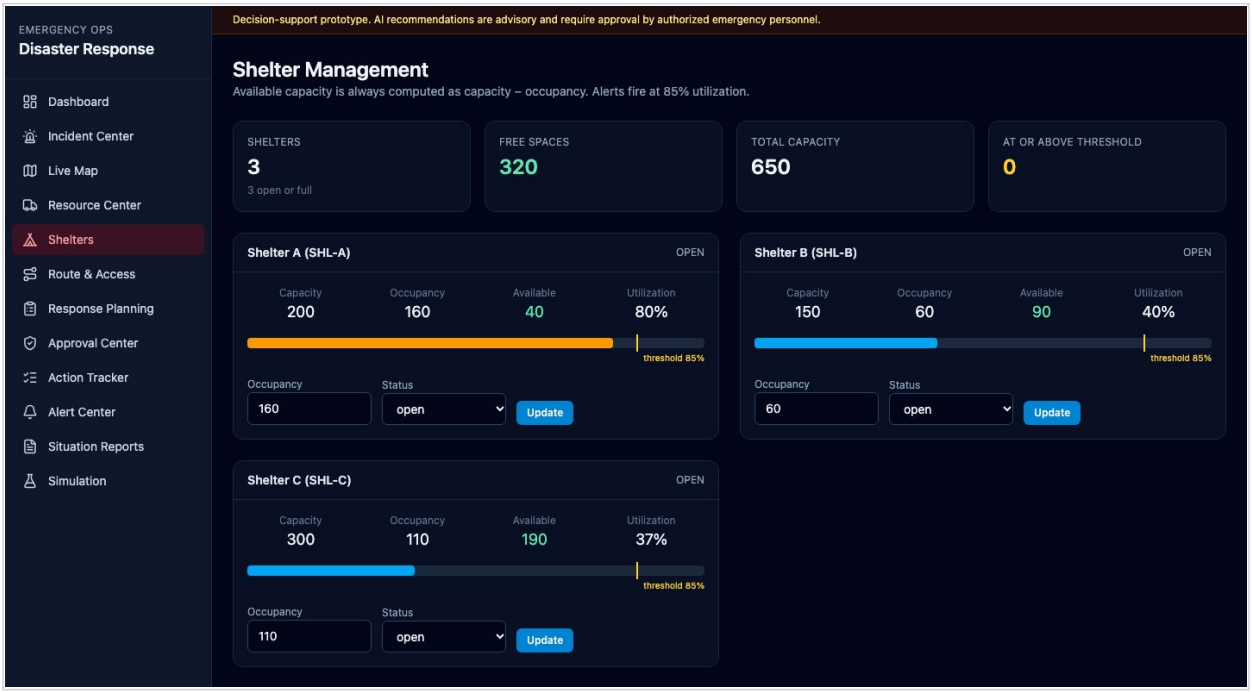


Figure 10.6: Shelter Management.

What you see

- Capacity, occupancy, **available** (always capacity minus occupancy) and utilisation, with a bar and the 85% alert threshold marker.

What you can do

- Update **occupancy** and **status** (open, full, closed) and click **Update**.
- Crossing 85% raises one warning alert; reaching full capacity sets the status to *full* and raises a critical alert.

10.7 Route & Access

Which zones can be reached by road, and how sure we are.

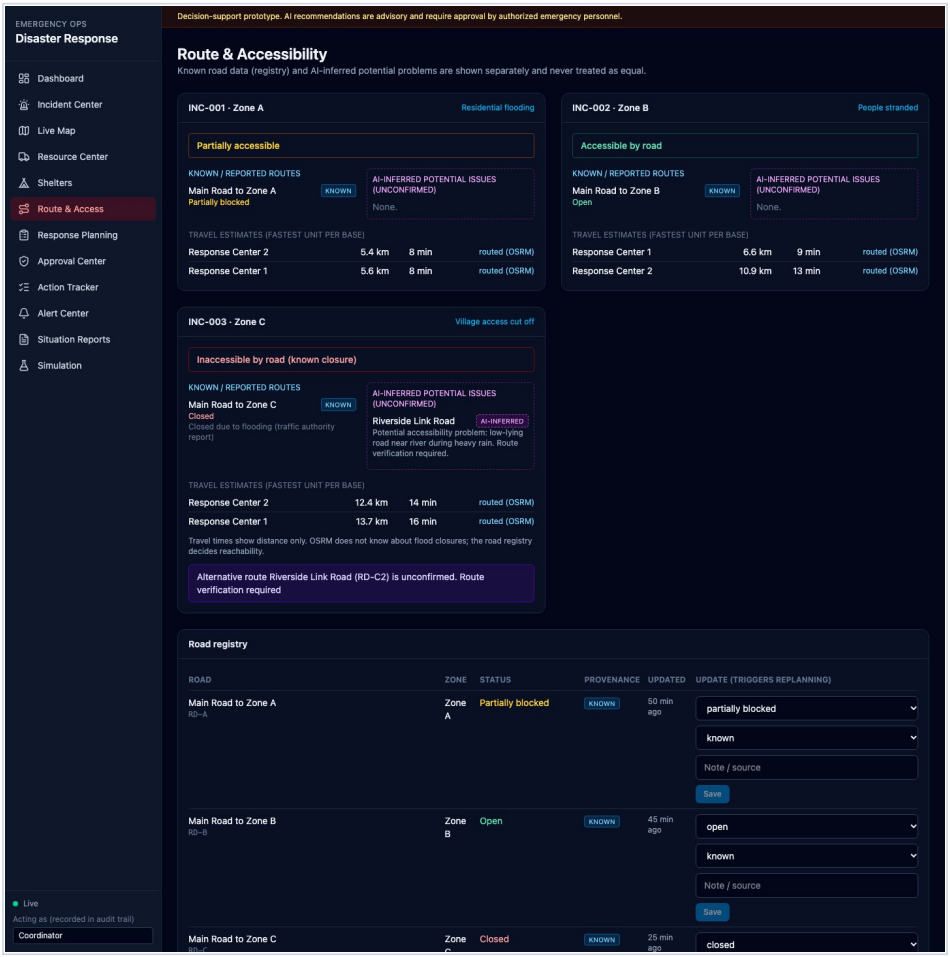


Figure 10.7: Route & Accessibility.

What you see

- Per incident: an access verdict (Accessible, Partially accessible, **Inaccessible by road (known closure)**, Unknown: route verification required).
- **Known / reported routes** and, separately in a dashed box, **AI-inferred potential issues (unconfirmed)**. They are never treated as equal.
- Travel estimates from each base, labelled *routed* (real road routing) or *estimated*.
- The **Road registry** table.

What you can do

- Change a road's **status** and **provenance**, add a note, and **Save**. This triggers replanning (chapter 12).

10.8 Response Planning

Compare the plans and see exactly how the agents reached them.

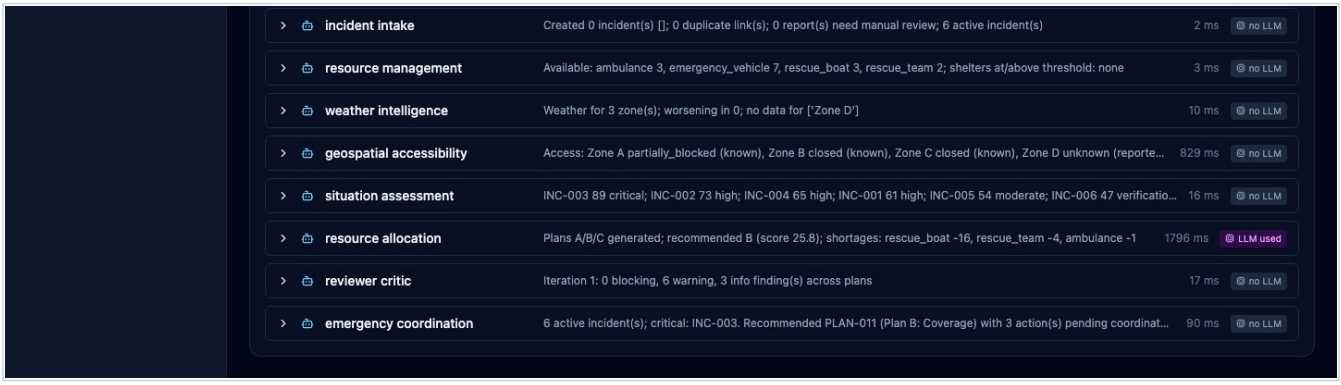


Figure 10.8: The agent trace at the bottom of Response Planning (one row per agent).

What you see

- Plans A, B and C side by side (Figure 8.1): explanation, per-incident allocations with a reason for every resource, unmet needs, assumptions, shortages and **Reviewer findings**.
- **Agent trace** of the latest run: each agent, its duration, whether the AI was used, and (when expanded) every tool call with arguments and results.

What you can do

- **Run assessment**: run all agents now and show the summary.
- Expand trace rows to audit the reasoning.

10.9 Approval Center

Where every recommendation waits for a person.

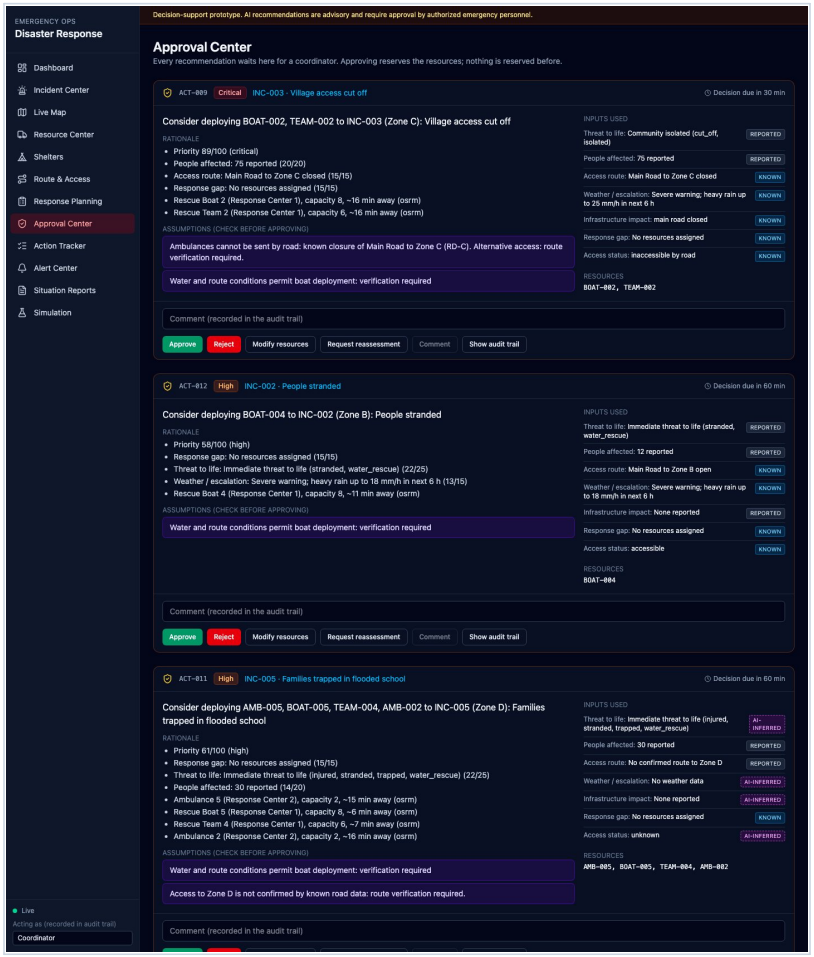


Figure 10.9: Approval Center cards.

What you see

- One card per pending action: priority, incident, the recommended resources and a **decision due** countdown (Critical 30 min, High 60, Moderate 120, Low 240).
- **Rationale** (why), **Assumptions (check before approving)** and **Inputs used** with their source badges.

What you can do

- **Approve**, **Reject**, **Modify resources** (tick available units by type), **Request reassessment**, **Comment**.
- **Show audit trail**: who did what and when.

> **Tip:** If you see "Cannot approve … BOAT-005 is reserved", someone else has just committed that resource. The card is replaced and a new recommendation appears within seconds.

10.10 Action Tracker

Follow every action to completion.

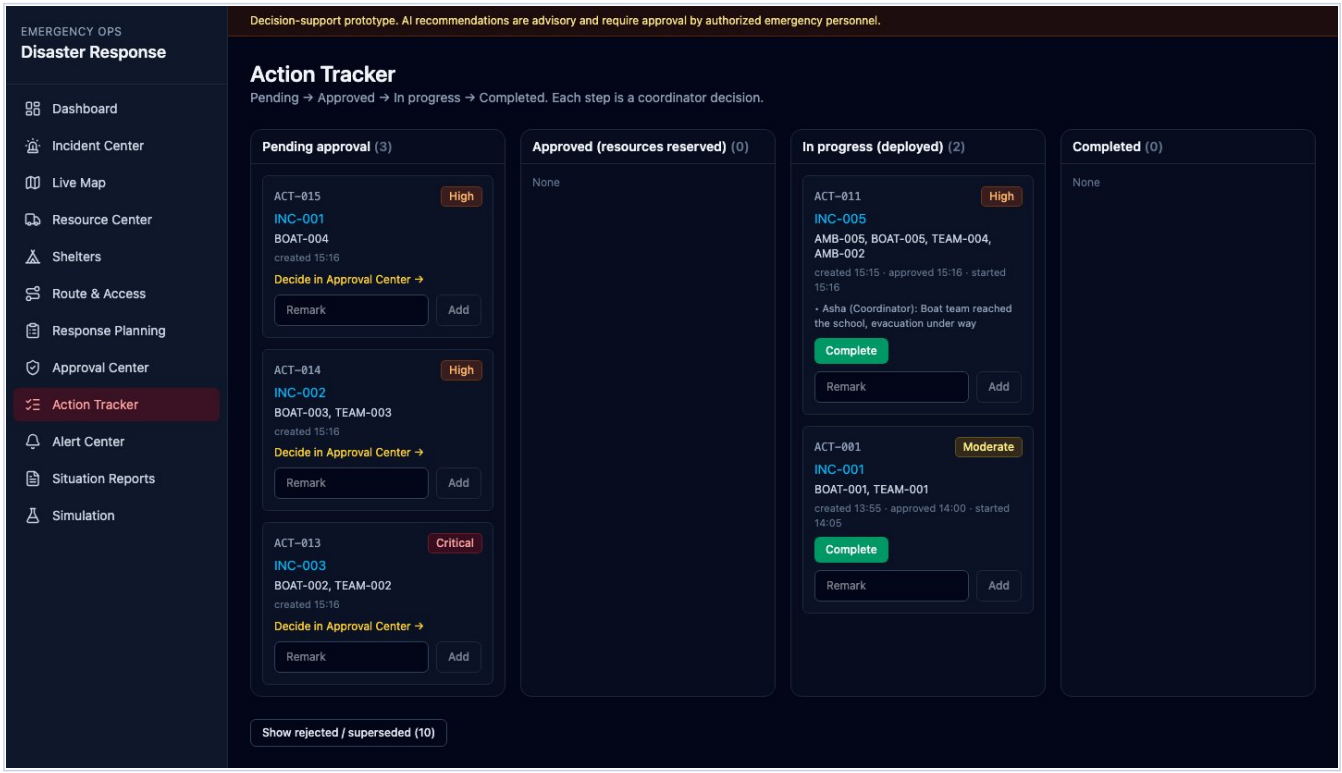


Figure 10.10: Action Tracker after a road closure.

What you see

- Four columns: **Pending approval**, **Approved (resources reserved)**, **In progress (deployed)**, **Completed**.
- Each card: resources, timestamps (created, approved, started, done), latest remarks; overdue cards are outlined in red.
- **Show rejected / superseded**: closed recommendations and why.

What you can do

- **Start** an approved action (resources become Deployed); **Complete** an in-progress one (resources become Returning).
- Add a **Remark** (recorded in the audit trail).

10.11 Alert Center

All alerts in one place.

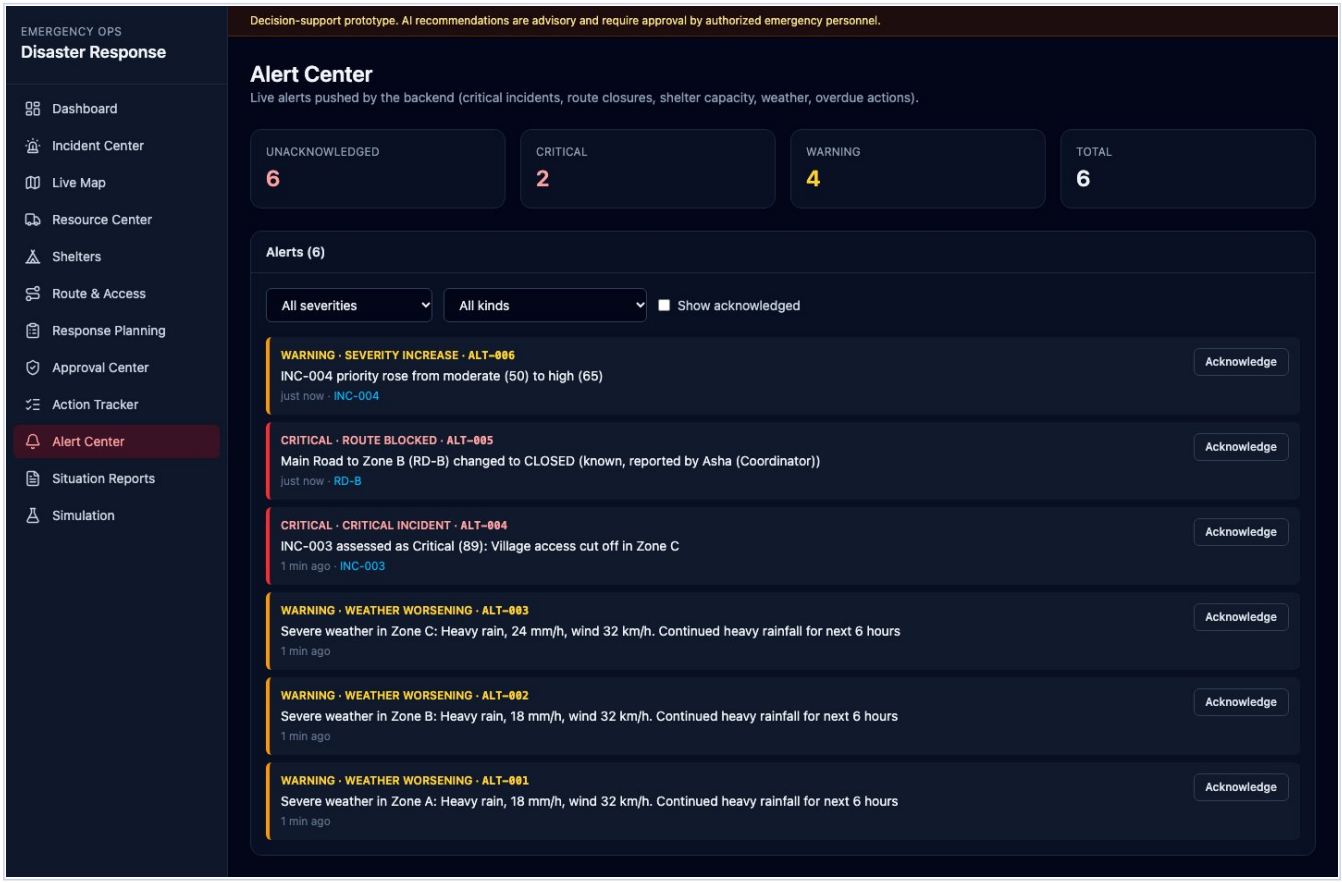


Figure 10.11: Alert Center.

What you see

- Counts of unacknowledged, critical and warning alerts; each alert with severity, kind, ID, message, time and a link to the related item.

What you can do

- Filter by **severity** and **kind**; **Show acknowledged**; **Acknowledge** an alert once handled.

10.12 Situation Reports

Shareable PDF reports of the current situation.

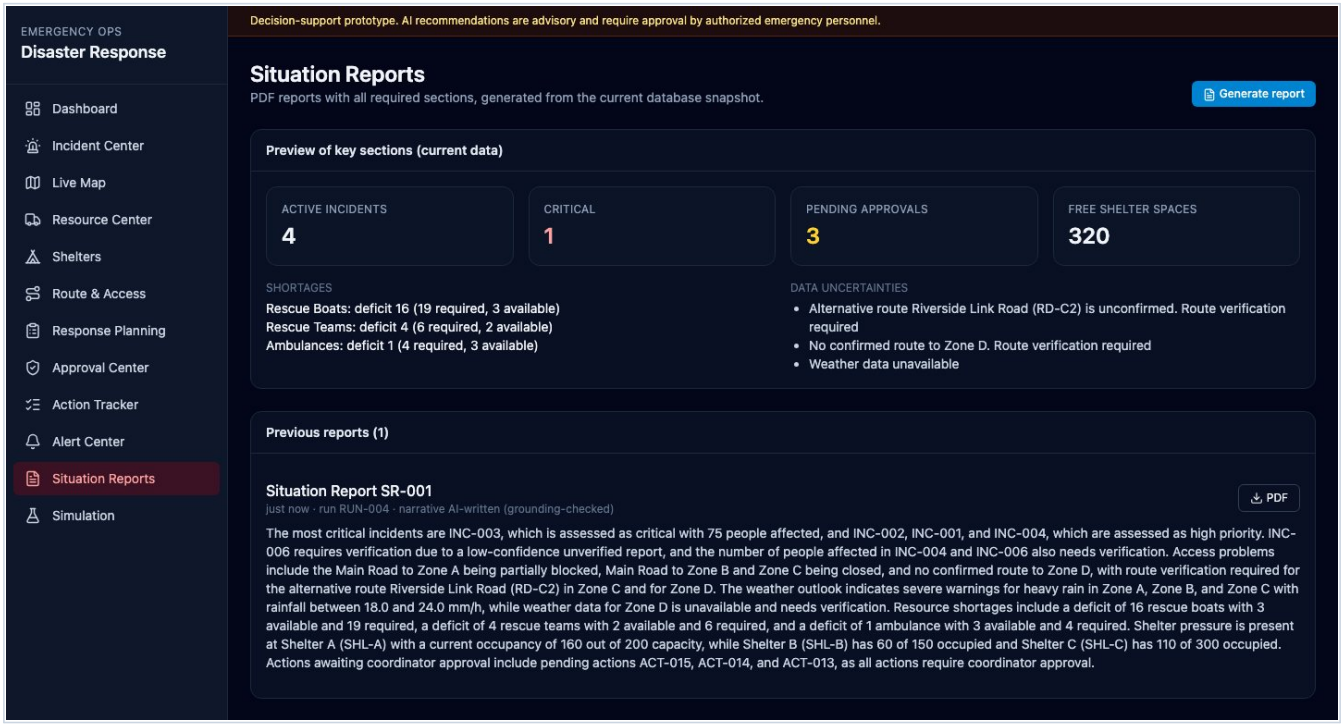


Figure 10.12: Situation Reports.

What you see

- A preview of key figures, shortages and data uncertainties.
- Previous reports with their narrative summary (written by the AI and fact-checked, or a template if the AI is unavailable).

What you can do

- **Generate report**, then **PDF** to download. Contents are listed in chapter 13.

10.13 Simulation (training and demonstrations)

Practise and demonstrate without waiting for real events.

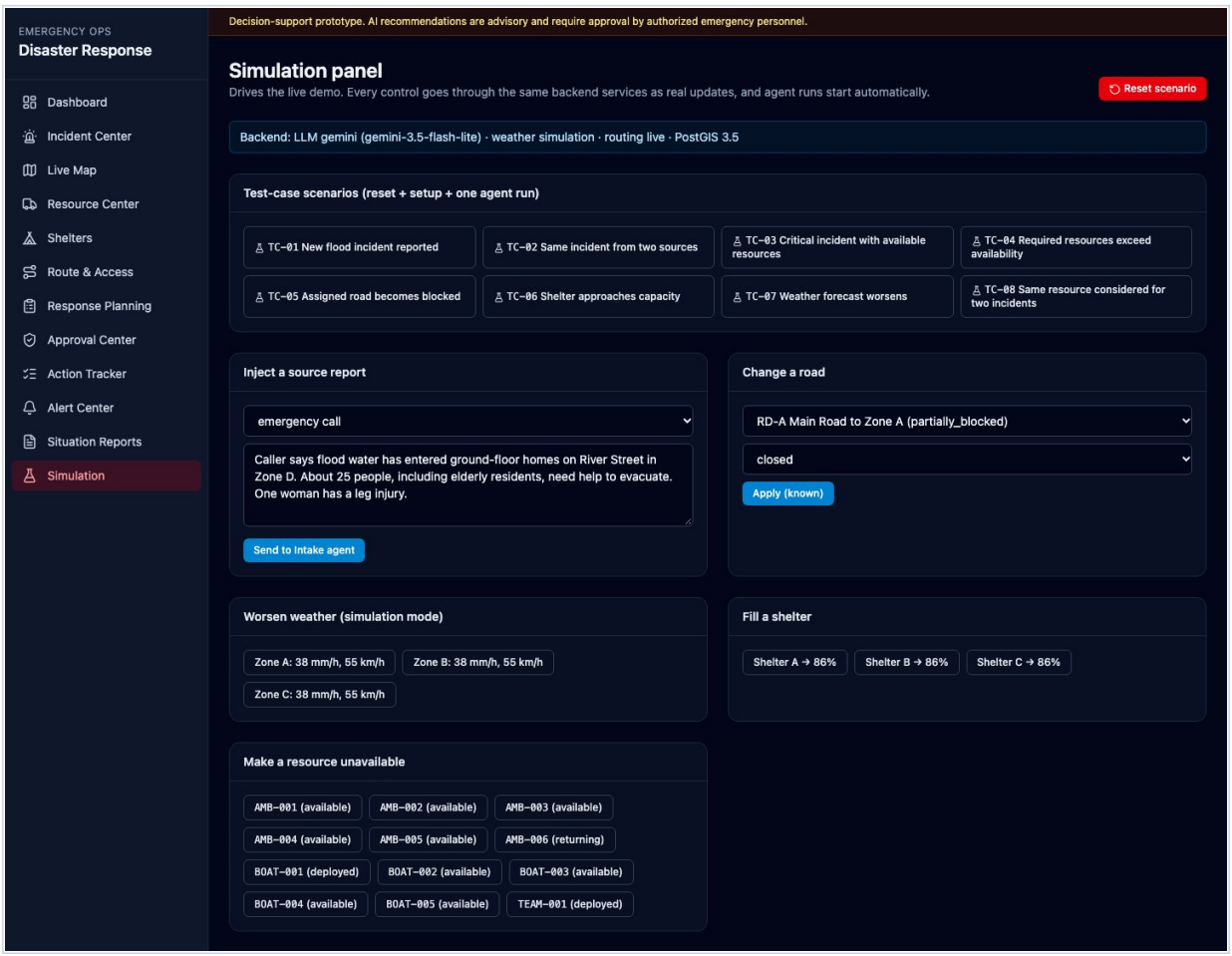


Figure 10.13: Simulation panel.

What you see

- Backend status (AI provider, weather and routing modes).
- Eight ready-made test scenarios (TC-01 to TC-08).

What you can do

- **Reset scenario**: reload the standard training data.
- **Test-case scenarios**: reset, set up a situation (new incident, duplicate reports, shortage, road closure, shelter pressure, worse weather, competing incidents) and run the agents.
- **Inject a source report** (any of five sources), **Change a road**, **Worsen weather**, **Fill a shelter**, **Make a resource unavailable**.

> **Tip:** Simulation controls change the shared training database. Do not use them on a live operational system.

11. Worked example: the flooded school

This is the complete story of one incident as it was run while preparing this manual.

#	What the user does / what happens	Where	Result
1	Load the training situation: Simulation → **TC-03**	Simulation	4 incidents; INC-003 Critical 89; 3 pending actions
2	Register *Families trapped in flooded school*, Zone D, field team, 30 people, water rescue + medical check	Incident Center → New incident	INC-005 created
3	AI reads the description: *trapped, injured, stranded, water rescue*	Incident Details	Threat to life 22/25 (AI-Inferred)
4	Agents run: no weather and no known road for Zone D → flagged, not guessed	Incident Details → Verification required	*Route verification required*, *Weather data unavailable*
5	Priority calculated	Incident Details	**61 / 100 High**
6	Plans built; boats are short (20 needed, 4 free); Plan B Coverage recommended	Response Planning	Shortage: rescue boats deficit 16
7	Pending action created for the school	Approval Center	ACT-011: AMB-005, BOAT-005, TEAM-004, AMB-002 (due in 60 min)
8	A citizen texts about the same school	(arrives automatically)	INC-006 created, linked as possible duplicate of INC-005 (0.79), not allocated separately
9	Coordinator approves ACT-011 with a comment	Approval Center	4 resources **Reserved**
10	Coordinator clicks **Start** and adds a remark	Action Tracker	4 resources **Deployed**; remark in audit trail
11	Traffic police close the Main Road to Zone B	Route & Access	Critical alert; old pending actions superseded; new run; new recommendations
12	Generate a situation report	Situation Reports	SR-001 PDF with AI-written, fact-checked narrative

The approval card the coordinator saw (ACT-011):

Part of the card	Content
Rationale	Priority 61/100 (high) · Response gap: no resources assigned (15/15) · Threat to life: immediate threat (22/25) · People affected: 30 reported (14/20) · Rescue Boat 5 (Response Center 1), capacity 8, ~6 min away · Rescue Team 4, capacity 6, ~7 min away · Ambulance 5 and Ambulance 2 (Response Center 2), ~15–16 min away
Assumptions	Water and route conditions permit boat deployment: verification required · Access to Zone D is not confirmed by known road data: route verification required
Inputs used	Threat to life (AI-Inferred) · People affected 30 (Reported) · Access route: no confirmed route (Reported) · Weather: no data (AI-Inferred) · Response gap (Known)
Audit trail	created by system → approved by Asha (Coordinator) → started by Asha (Coordinator) → comment: "Boat team reached the school, evacuation under way"

12. When conditions change

The console keeps the plan current. Any of these changes starts the replanning process automatically:

Change	How it is entered	Alert raised
Road closed or partly blocked	Route & Access → road registry (or Simulation)	Route blocked: critical (closed) or warning (partly blocked)
Weather gets worse	Automatically from the weather service (or Simulation)	Weather worsening (warning)
Resource breaks down	Resource Center → Change status → unavailable	Resource unavailable: critical if it was assigned
Shelter fills up	Shelters → Update occupancy	Shelter capacity: warning at 85%, critical when full
New information about an incident	Incident Details, a new report, or a people-count update	Severity increase if the level rises; Critical incident if it becomes critical
Action not decided or started in time	Automatic check every minute	Action overdue (warning)

12.1 What replanning does

1. Records the change (old value → new value, with its source) in *Recently changed conditions*.
2. Works out which incidents and actions are affected.
3. Replaces affected **pending** recommendations (they move to *superseded* with a note explaining why).
4. Adds a **coordinator review required** note to affected **approved or in-progress** actions. It never changes them by itself.
5. Raises the alert and pops up a notification on every open screen.
6. Runs all eight agents again and publishes the new recommendations in the Approval Center.

In the worked example, closing the Main Road to Zone B produced a critical *route blocked* alert, raised INC-004 from Moderate (50) to High (65) with a *severity increase* alert, superseded four pending actions and created three new ones (Figure 10.10).

13. Situation reports

Click **Situation Reports → Generate report**, then **PDF**. Every report is built from the live data at that moment and contains these sections, with the safety disclaimer on every page:

#	Section	#	Section
1	Situation summary (narrative)	9	Resource shortages
2	Incident summary	10	Shelter status
3	Affected locations	11	Recommended actions
4	Priority assessment	12	Pending approvals
5	Weather conditions	13	Approved, in-progress and completed actions
6	Accessibility (known vs inferred)	14	Data uncertainties
7	Available resources	15	Assessment timestamp
8	Deployed / reserved / returning resources		

Note: The narrative is written by the AI only from the report's own data and is automatically fact-checked: if it mentions any ID or number that is not in the data, it is replaced by a standard template. The report states which one was used.

14. Troubleshooting and FAQ

Question or message	What it means and what to do
Cannot reach the backend at … or the sidebar dot is red (Offline)	The server is not running or not reachable. Ask your administrator to start it; the screen reconnects automatically.
The incident says *Not yet assessed*	The agents are still running. Wait a few seconds; the page updates by itself. If not, click **Reassess now**.
Why is the level **Verification Required**?	The location is unknown, or a low-confidence unverified report is not confirmed. Verify the incident or confirm a duplicate link.
Why didn't the system send an ambulance?	Road-only units are never sent through a known road closure. The plan shows the unmet need and the reason.
Why is a resource missing from the plans?	It is not *available* (reserved, deployed, returning or unavailable), or it was already used by a higher-priority incident in that plan.
Cannot approve … is reserved (conflict)	Another approval committed that resource first. The card is replaced and a fresh plan appears shortly.
A pending action disappeared	Conditions changed and it was **superseded** by a newer recommendation. See *Show rejected / superseded* in the Action Tracker.
The trace shows *fallback* instead of *LLM used*	The AI service was unavailable; the rule-based reader was used and results are marked low confidence. Work can continue.
The weather says *simulation*	The console is in training mode with scripted weather. Live mode uses real weather data.
Shortage numbers look large	They are exact: everything needed by all incidents minus what is free. Use the escalation note to request more capacity.

15. Glossary

Term	Meaning
Agent	A specialised AI component with one job (intake, weather, access, resources, severity, planning, review, coordination).
Audit trail	The permanent list of who did what to an action, and when.
Coverage	Plan B's strategy: a minimum response for every serious incident first.
Deficit	Required minus available for a resource type.
Grounding check	Automatic fact-check that AI-written text only mentions IDs and numbers present in the data.
Provenance	Where a fact came from: Known, Reported or AI-Inferred.
Run	One complete pass of all eight agents.
Superseded	A pending recommendation replaced by a newer one after conditions changed.
Verification Required	The system will not rely on this fact until a person confirms it.